Data Encapsulation

3.6.1

Segmenting Messages

Knowing the OSI reference model and the TCP/IP protocol model will come in handy when you learn about how data is encapsulated as it moves across a network. It is not as simple as a physical letter being sent through the mail system.

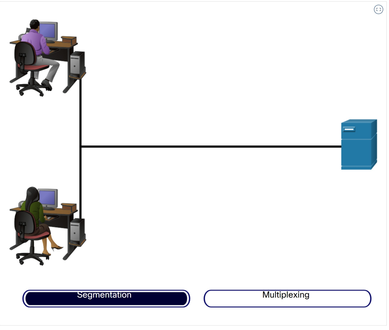
In theory, a single communication, such as a video or an email message with many large attachments, could be sent across a network from a source to a destination as one massive, uninterrupted stream of bits. However, this would create problems for other devices needing to use the same communication channels or links. These large streams of data would result in significant delays. Further, if any link in the interconnected network infrastructure failed during the transmission, the complete message would be lost and would have to be retransmitted in full.

A better approach is to divide the data into smaller, more manageable pieces to send over the network. Segmentation is the process of dividing a stream of data into smaller units for transmissions over the network. Segmentation is necessary because data networks use the TCP/IP protocol suite send data in individual IP packets. Each packet is sent separately, similar to sending a long letter as a series of individual postcards. Packets containing segments for the same destination can be sent over different paths.

This leads to segmenting messages having two primary benefits:

- **Increases speed** – Because a large data stream is segmented into packets, large amounts of data can be sent over the network without tying up a communications link. This allows many different conversations to be interleaved on the network called multiplexing.
- **Increases efficiency** –If a single segment is fails to reach its destination due to a failure in the network or network congestion, only that segment needs to be retransmitted instead of resending the entire data stream.

Click each button in the figure to view the animations of segmentation and multiplexing.

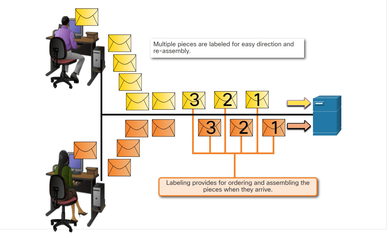


3.6.2

Sequencing

The challenge to using segmentation and multiplexing to transmit messages across a network is the level of complexity that is added to the process. Imagine if you had to send a 100-page letter, but each envelope could only hold one page. Therefore, 100 envelopes would be required and each envelope would need to be addressed individually. It is possible that the 100-page letter in 100 different envelopes arrives out-of-order. Consequently, the information in the envelope would need to include a sequence number to ensure that the receiver could reassemble the pages in the proper order.

In network communications, each segment of the message must go through a similar process to ensure that it gets to the correct destination and can be reassembled into the content of the original message, as shown in the figure. TCP is responsible for sequencing the individual segments.



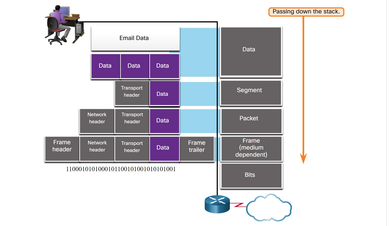
3.6.3

Protocol Data Units

As application data is passed down the protocol stack on its way to be transmitted across the network media, various protocol information is added at each level. This is known as the encapsulation process.

Note: Although the UDP PDU is called datagram, IP packets are sometimes also referred to as IP datagrams.

The form that a piece of data takes at any layer is called a protocol data unit (PDU). During encapsulation, each succeeding layer encapsulates the PDU that it receives from the layer above in accordance with the protocol being used. At each stage of the process, a PDU has a different name to reflect its new functions. Although there is no universal naming convention for PDUs, in this course, the PDUs are named according to the protocols of the TCP/IP suite. The PDUs for each form of data are shown in the figure.



- Data – The general term for the PDU used at the application layer
- Segment – Transport layer PDU
- Packet – Network layer PDU
- Frame – Data Link layer PDU
- Bits – Physical layer PDU used when physically transmitting data over the medium

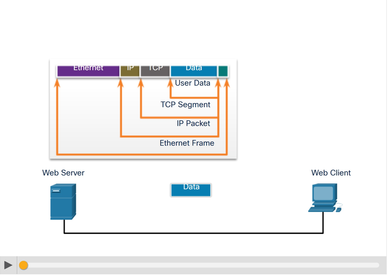
Note: If the Transport header is TCP, then it is a segment. If the Transport header is UDP then it is a datagram.

3.6.4

Encapsulation Example

When messages are being sent on a network, the encapsulation process works from top to bottom. At each layer, the upper layer information is considered data within the encapsulated protocol. For example, the TCP segment is considered data within the IP packet.

You saw this animation previously in this module. This time, click Play and focus on the encapsulation process as a web server sends a web page to a web client.

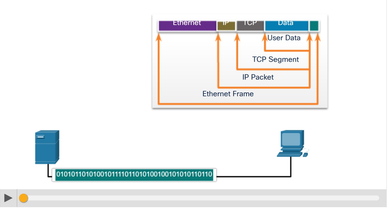


3.6.5

De-encapsulation Example

This process is reversed at the receiving host and is known as de-encapsulation. De-encapsulation is the process used by a receiving device to remove one or more of the protocol headers. The data is de-encapsulated as it moves up the stack toward the end-user application.

You saw this animation previously in this module. This time, click Play and focus on the de-encapsulation process.



3.6.6

Check Your Understanding - Data Encapsulation

Check your understanding of data encapsulation by choosing the BEST answer to the following questions.

1. What is the process of dividing a large data stream into smaller pieces prior to transmission?
   - sequencing
   - duplexing
   - multiplexing
   - segmentation
2. What is the PDU associated with the transport layer?
   - segment
   - packet
   - bits
   - frame
3. Which protocol stack layer encapsulates data into frames?
   - data link
   - transport
   - network
   - application
4. What is the name of the process of adding protocol information to data as it moves down the protocol stack?
   - de-encapsulation
   - sequencing
   - segmentation
   - encapsulation

Check

Show Me

Reset